HELICOPTER ENGINEERING

PROFESSIONAL ELECTIVES-III

VII Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6AE42	PEC	3	0	0	3	40	60	100

UNIT-I ROTORCRAFT AND ROTOR CONTROL

TYPES OF ROTORCRAFTS: Helicopter - main rotor system, Features of Fully Articulated Rotor System, Features of Semi Rigid Rotor System, Features of Rigid Rotor System. Transmission system - Main Rotor drive system, Tail Rotor Drive System. Configurations based on torque reaction, Jet rotors and compound helicopters.
ROTOR CONTROL: Methods of control, Collective and cyclic pitch changes, Lead-lag and flapping hinges

UNIT-II IDEAL ROTOR THEORY AND ROTOR PERFORMANCE

IDEAL ROTOR THEORY: Hovering performances, Momentum and simple blade element theories.
ROTOR PERFORMANCE: Figures of merit, Profile and induced power estimation, Constant chord and ideal twist rotors.

UNIT-III POWER ESTIMATES AND STABILITY AND TRIM

POWER ESTIMATES: Induced, Profile and Parasite power requirements in forward flight, Performances curves with effects of altitude.
STABILITY AND TRIM: Hover Trim, Trim in Forward Flight, Damping – Static Stability, Static Longitudinal Stability, Angle of Attack Stability, Static Directional Stability.

UNIT-IV LIFT AND CONTROL OF V/S TOL AIRCRAFT

Various configuration, Propeller, Rotor ducted fan and jet lift, Tilt wing and vectored thrust, Performances of VTOL and STOL aircraft in hover, Transition and Forward motion

UNIT-V GROUND EFFECT MACHINES

Types, Hover height, Lift augmentation and power calculations for plenum chamber and peripheral jet machines. Drag of hovercraft on land and water. Applications of hovercraft

Text Books:

1. Johnson Wayne (2011), *Helicopter Theory*, 1st edition, Sterling Publishing House, NewYork.
2. McCormick B. W. (2010), *Aerodynamics, Aeronautics and Flight Mechanics,* 2nd edition, Wiley India Ltd, New Delhi,India

Reference Books:

1. Alfred Gessow, Garry C. Myers (2007), Aerodynamics of Helicopter, 2nd edition, F. Ungar Pub. Co, New York.
2. B.W. (1998), Aerodynamics of V/STOL Flight, Dover Publications,USA.
3. John M. Seddon (2011), Basic Helicopter Aerodynamics, John Wiley & Sons,USA.

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Determine the working characteristics of rotors and flight controllers for flights
2. Derive the blade element theory and profile induced power estimation.
3. Derive the stability and coefficient of performance of forward flight
4. Determine the performance analysis of VTOL and STOL aircraft.
5. Relate the power requirement for jet machines and drag of hovercraft on land and water.